

FULL NAME:

NETID:

CSC 380 Sample Midterm Exam

- *Exam Duration: 75 minutes.*
- *There are 6 questions.*
- *Provide your answers **only inside the boxes** underneath the questions.*
- *Provide neat and concise solutions.*
- *Good luck!*

Question 1 : Mark TRUE or FALSE and briefly explain with one or two sentences. Note that you will have points cut off if your answer is correct, but your explanation is not or if your explanation is missing.

- (a) Sample mean is an unbiased estimator for probability of heads p of a Bernoulli distribution.

TRUE FALSE

- (b) $\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2$ is an unbiased estimator of variance.

TRUE FALSE

- (c) Let $X_1, \dots, X_n \sim \text{Bernoulli}(\theta)$. The maximum likelihood estimator for unknown θ is the same as the sample mean.

TRUE FALSE

- (d) Weak law of large numbers claims that as the size of sample increases, the sample mean converges to the distribution mean in probability.

TRUE FALSE

- (e) The quantile function is the inverse of the pdf.

TRUE FALSE

- (f) Sampling n balls with replacement from a box containing R red balls and B blue balls can be modeled with a hypergeometric distribution.

TRUE FALSE

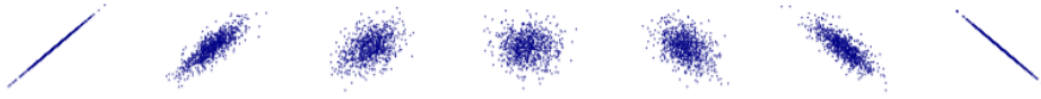
Question 2 :

The exponential distribution has cdf $1 - e^{-\lambda x}$. Describe how to sample two random points from this distribution using inverse transform sampling.

Answer:

Question 3 :

- (a) Associate each plot below with one of the Pearson correlation coefficient values: -0.4, 0.8, -0.8, -1, 1, 0, 0.4.



Answer:

- (b) Discuss briefly what the phrase *correlation does not imply causation* means and provide an example.

Answer:

- (c) Discuss what Z-score normalization is and why it is useful in handling data compatibility issues while doing data cleaning.

Answer:

- (d) Describe why we need multiple testing correction and a method to resolve it.

Answer:

Question 4 :

- (a) Suppose that a baseball hitter has a probability of success $p = 0.7$. What is the probability that she hits (has success) more than 6 times out of a total of 15 throws. You may make use of the following outputs from `scipy.stats`: $\text{binom.pmf}(5, 15, 0.7) = 0.003$, $\text{binom.cdf}(5, 15, 0.7) = 0.004$, $\text{binom.pmf}(6, 15, 0.7) = 0.012$, $\text{binom.cdf}(6, 15, 0.7) = 0.015$.

Answer:

- (b) Suppose that X has the normal distribution with mean 5 and standard deviation 2. Find out the value of $P(1 < X < 8)$. You may make use of the following outputs from `scipy.stats`: $\text{norm.pdf}(2) = 0.054$, $\text{norm.cdf}(2) = 0.977$, $\text{norm.pdf}(3) = 0.004$, $\text{norm.cdf}(1.5) = 0.933$.

Answer:

Question 5 :

- (a) Two boxes contain long bolts and short bolts. Suppose that one box contains 60 long bolts and 40 short bolts, and that the other box contains 10 long bolts and 20 short bolts. Suppose also that one box is selected at random and a bolt is then selected at random from that box. What is the probability that this bolt is long?

Answer:

- (b) Suppose that X and Y are random variables such that $Var(X) = 9$, $Var(Y) = 4$, and the Pearson correlation coefficient of the two variables $r = -1/6$. Determine $Var(X + Y)$.

Answer:

Question 6 :

- (a) We take a random sample of size 64 from a distribution with unknown mean μ that has standard deviation $\sigma = 4$. Assume the sample mean is 6. What is the 90% confidence interval for mean μ ? You may make use of the following outputs from `scipy.stats`: $\text{norm.ppf}(0.995) = 2.58$, $\text{norm.ppf}(0.95) = 1.64$, $\text{norm.cdf}(0.995) = 0.84$, $\text{norm.cdf}(0.95) = 0.83$.

Answer:

- (b) We take a random sample of size 4 from a distribution with unknown mean μ and unknown σ . The values of the sample are 3, 5, 2, 6. What is the 90% confidence interval for mean μ ? You may make use of the following outputs from `scipy.stats`: $\text{norm.ppf}(0.95) = 1.64$, $t.ppf(0.95, 3) = 2.35$, $\text{norm.ppf}(0.995) = 2.58$, $t.ppf(0.995, 3) = 5.84$.

Answer: